

Yong-Gyu Choi

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Last update: 10th April, 2026

Education

- 2019–2025 **Ph.D. in Mathematics**, KAIST, Daejeon
Advisor: Prof. Wansu Kim.
Dissertation: “On degenerations of D^* -shtukas over ramified legs”
2015–2019 **B.S. in Mathematics**, KAIST, Daejeon, *magna cum laude*

Experience

- 2025– **Postdoctoral Fellow**, UNIST, Ulsan

Research Interests

- Arithmetic geometry
- Algebraic number theory
- Moduli stacks of shtukas and Shimura varieties, and their arithmetic
- Langlands program
- Trace formula

Award

- 2019 **Best Teaching Assistant Awards**, Spring 2019, Department of Mathematical Sciences, KAIST

Preprint

- 2025 Y.-G. Choi, Wansu Kim, and Junyeong Park, *On properness of moduli stacks of $D^\times$ -shtukas over ramified legs*, arXiv:2505.18977

Talks

- 2026 *Degenerations of D -shtukas over ramified legs*, France-Korea Webinar in Number Theory, April 2026
Degenerations of D -shtukas over ramified legs, Workshop on Automorphic forms, Arithmetic, and Geometry (WAAG-I), Korea Institute of Advanced Study, March 2026
Degenerations of D -shtukas over ramified legs, 2026 Annual Number Theory Conference, Pusan National University, February 2026
2025 *Degenerations of $D^\times$ -shtukas over ramified legs*, G-BRL Number Theory Seminar at CNU, Chonnam National University, May 2025
Degenerations of $D^\times$ -shtukas over ramified legs, PMI Number Theory Seminar, Postech, May 2025

Degenerations of $\mathcal{D}^\times$ -shtukas over ramified legs, 2025 KMS(Korean Mathematical Society) Spring Meeting, KAIST, April 2025

2023 *On Degenerations of D -shtukas over ramified legs*, Mini-workshop for young number theorists, KIAS, November 2023

2022 *Introduction to prismatic cohomology*, 2022 UNIST Summer Workshop on Langlands Program, UNIST, July 2022

■ Teaching assistantships

(All of my teaching assistantships were at KAIST.)

- Fall 2023 Calculus II (MAS102) and Linear Algebra (MAS212)
- Spring 2023 Calculus II (MAS102) and Linear Algebra (MAS212)
- Fall 2022 Calculus I (MAS101) and Calculus II (MAS102)
- Spring 2022 Modern Algebra (MAS311)
- Fall 2021 Calculus II (MAS102) and Linear Algebra (MAS212)
- Spring 2021 Calculus I (MAS101) and Calculus II (MAS102)
- Fall 2020 Introduction to Algebraic Geometry (MAS441)
- Spring 2020 Introduction to Linear Algebra (MAS109) and Algebra I (MAS511)
- Fall 2019 Calculus II (MAS102)
- Spring 2019 Calculus I (MAS101)